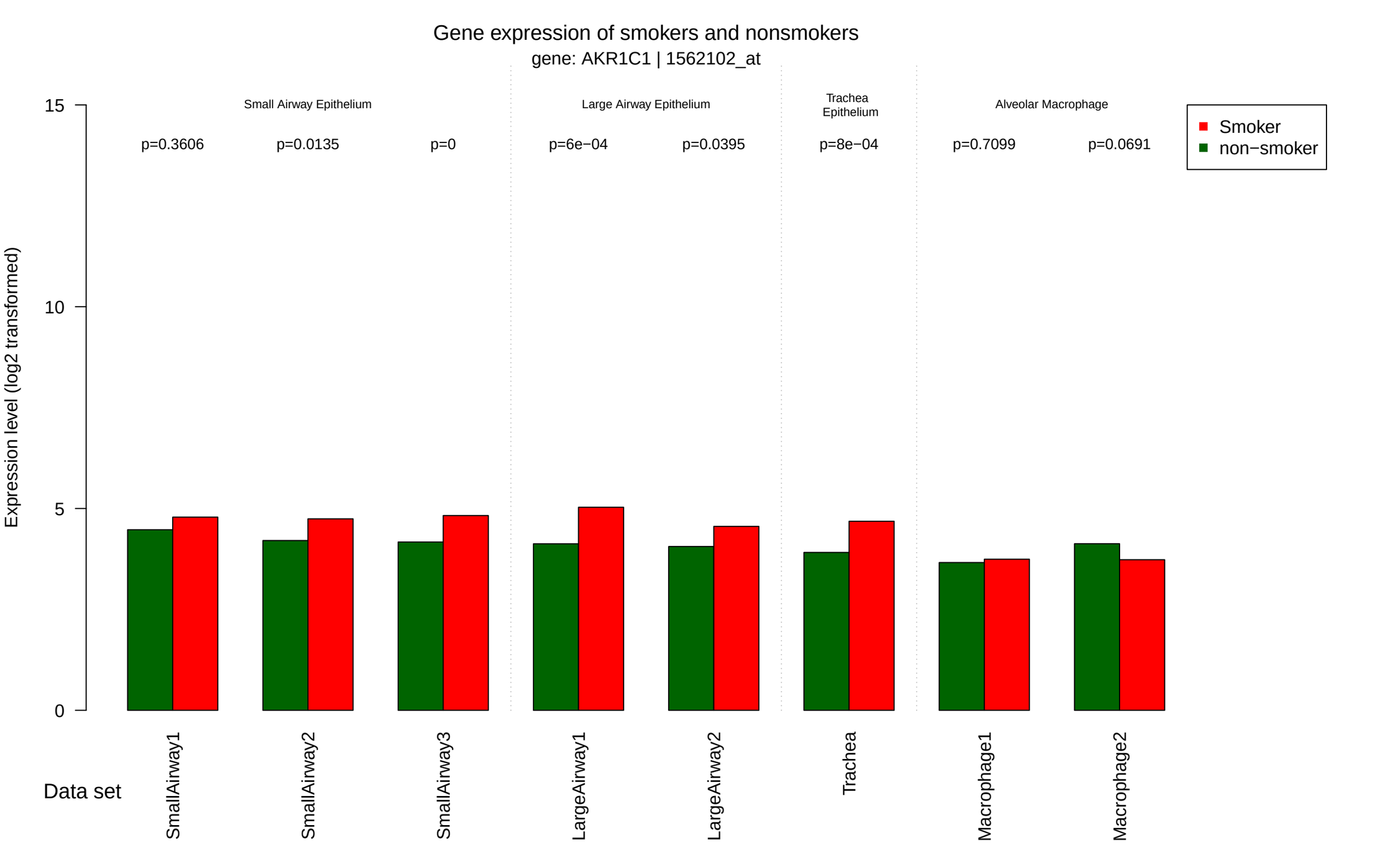
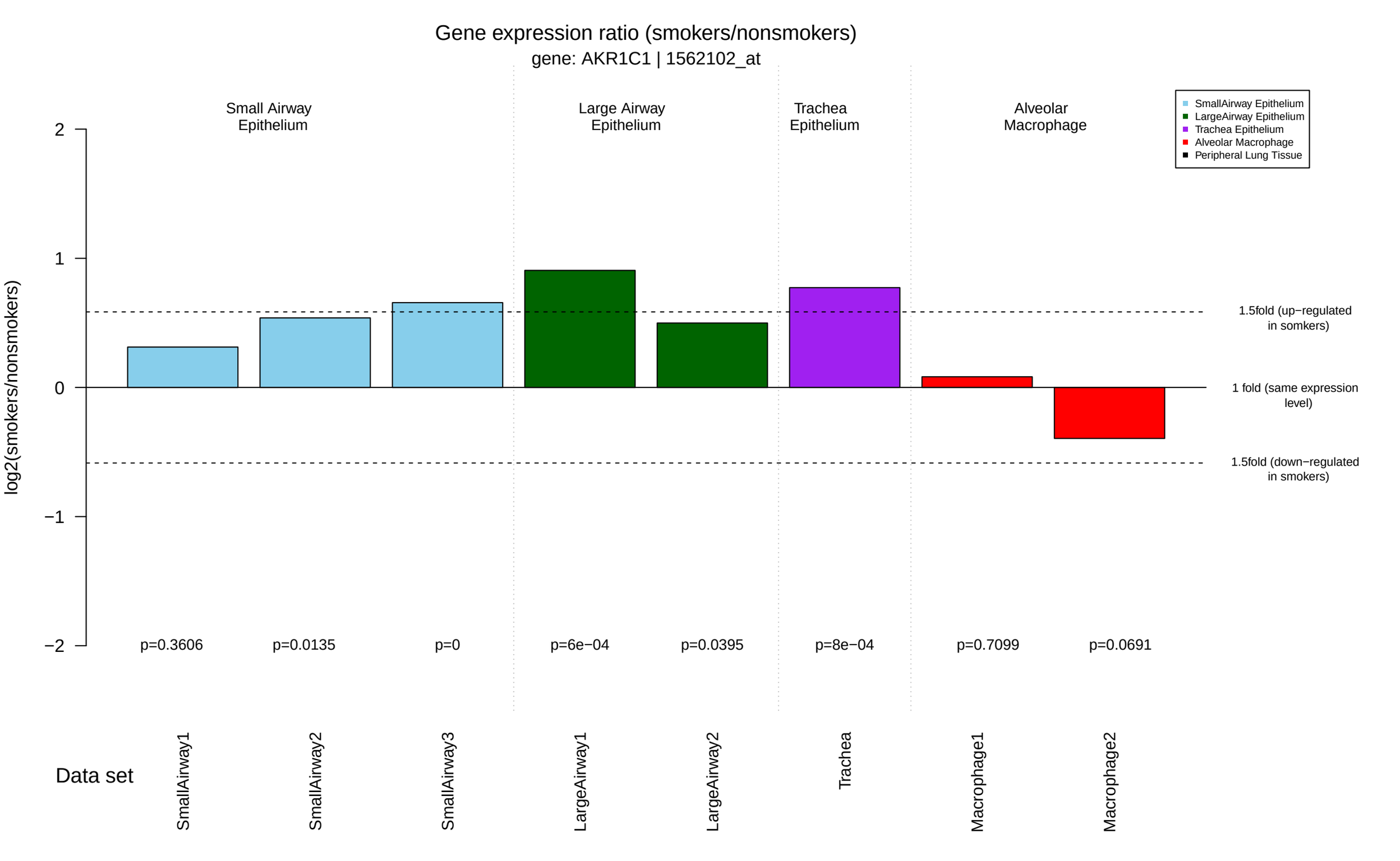
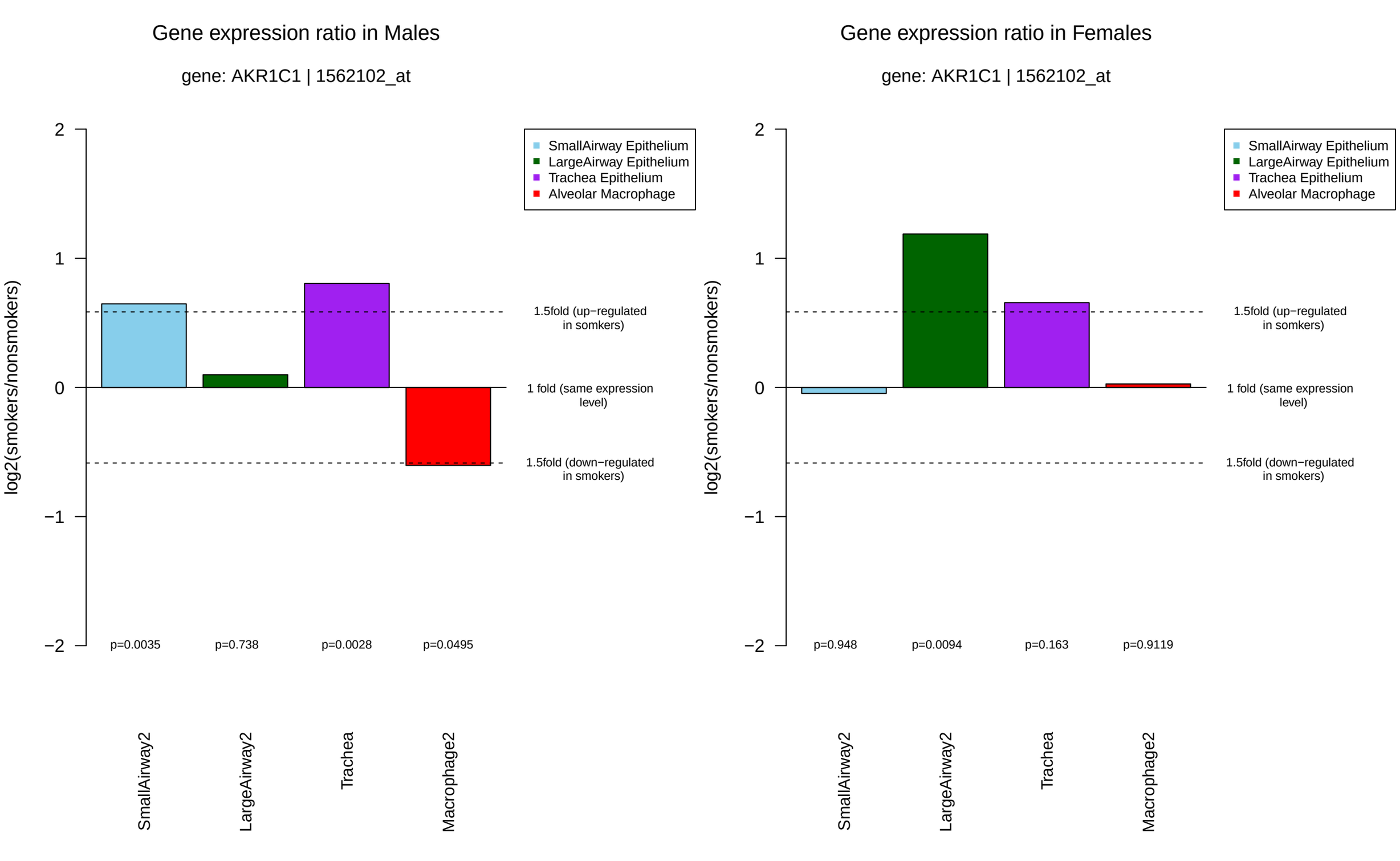
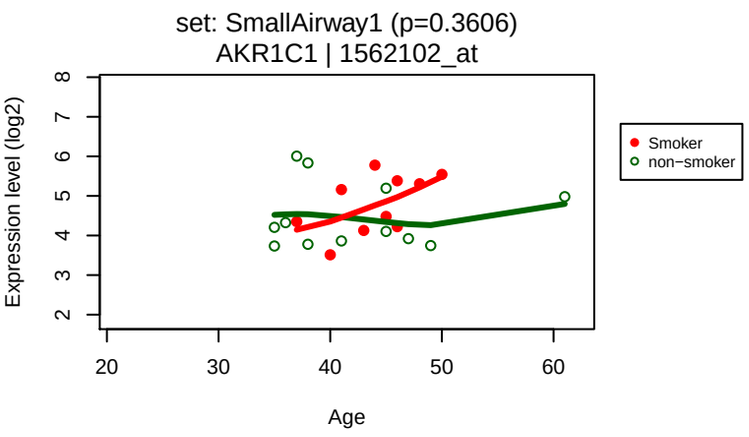
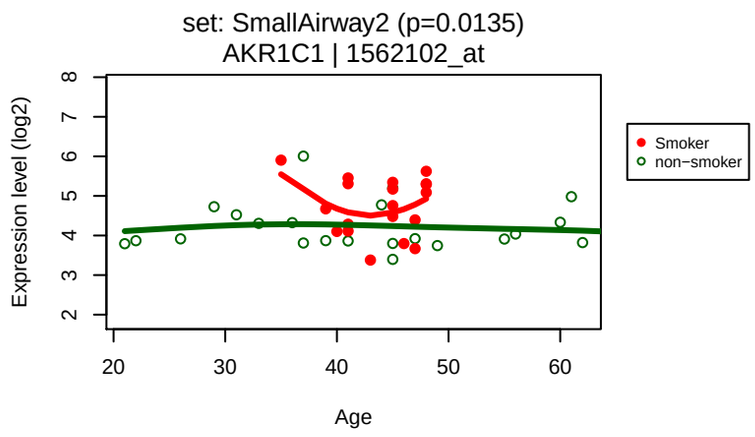
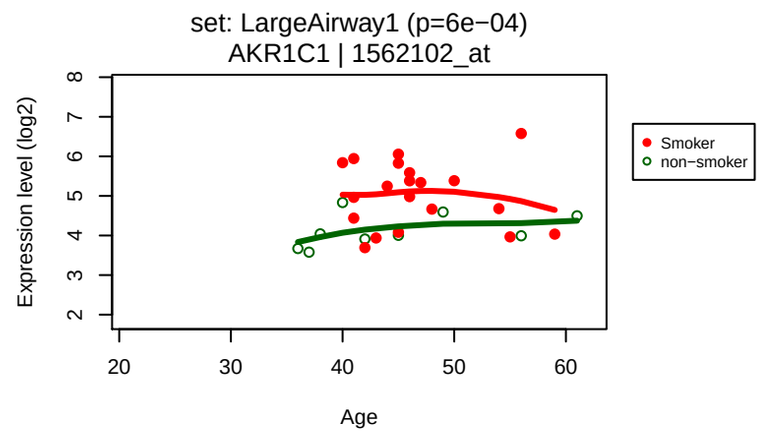

Gene expression of smokers and nonsmokers
gene: AKR1C1 | 1562102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3606
p=0.0135
p=0
p=6e−04
p=0.0395
p=8e−04
p=0.7099
p=0.0691
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

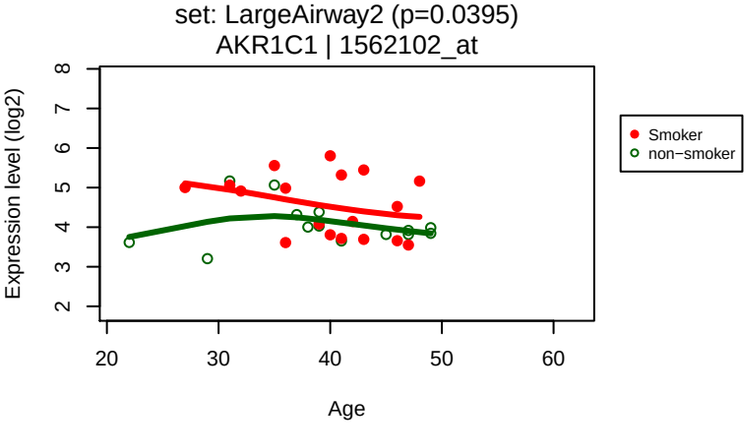

Gene expression ratio (smokers/nonsmokers)
gene: AKR1C1 | 1562102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3606
p=0.0135
p=0
p=6e−04
p=0.0395
p=8e−04
p=0.7099
p=0.0691
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

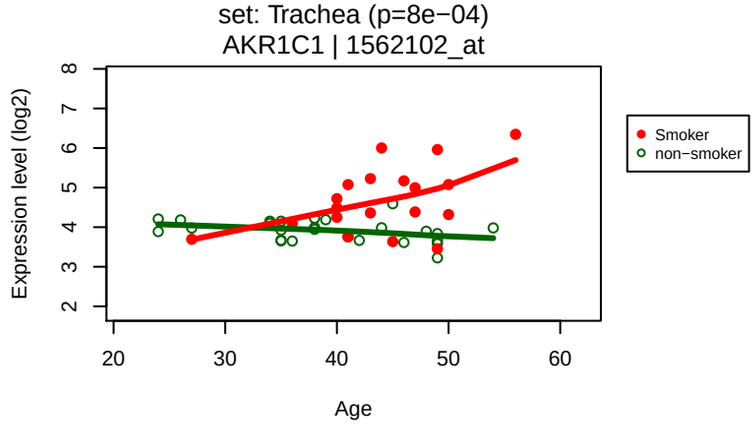

Gene expression ratio in Males
gene: AKR1C1 | 1562102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0035
p=0.738
p=0.0028
p=0.0495
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR1C1 | 1562102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.948
p=0.0094
p=0.163
p=0.9119
SmallAirway2
LargeAirway2
Trachea
Macrophage2

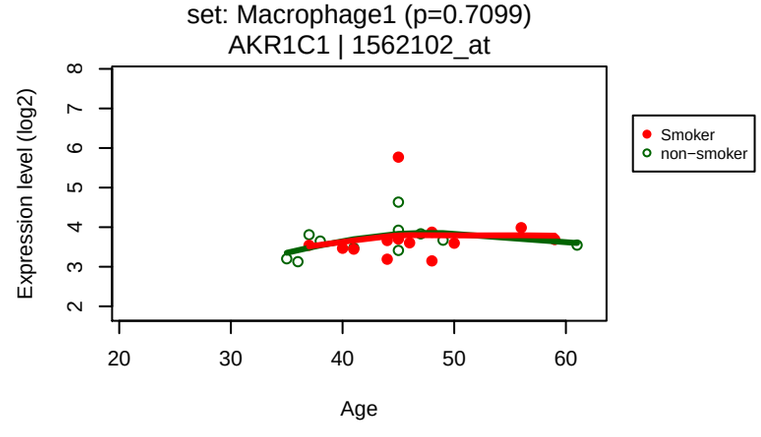

set: SmallAirway1 (p=0.3606)
AKR1C1 | 1562102_at
Expression level (log2)
Age
Smoker
non−smoker

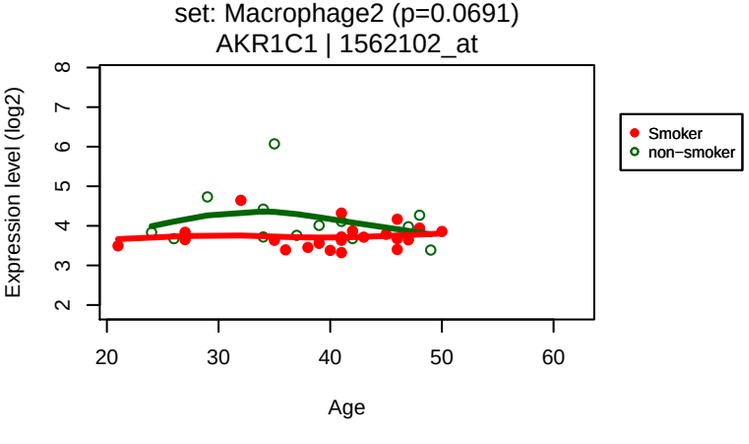

set: SmallAirway2 (p=0.0135)
AKR1C1 | 1562102_at
Expression level (log2)
Age
Smoker
non−smoker

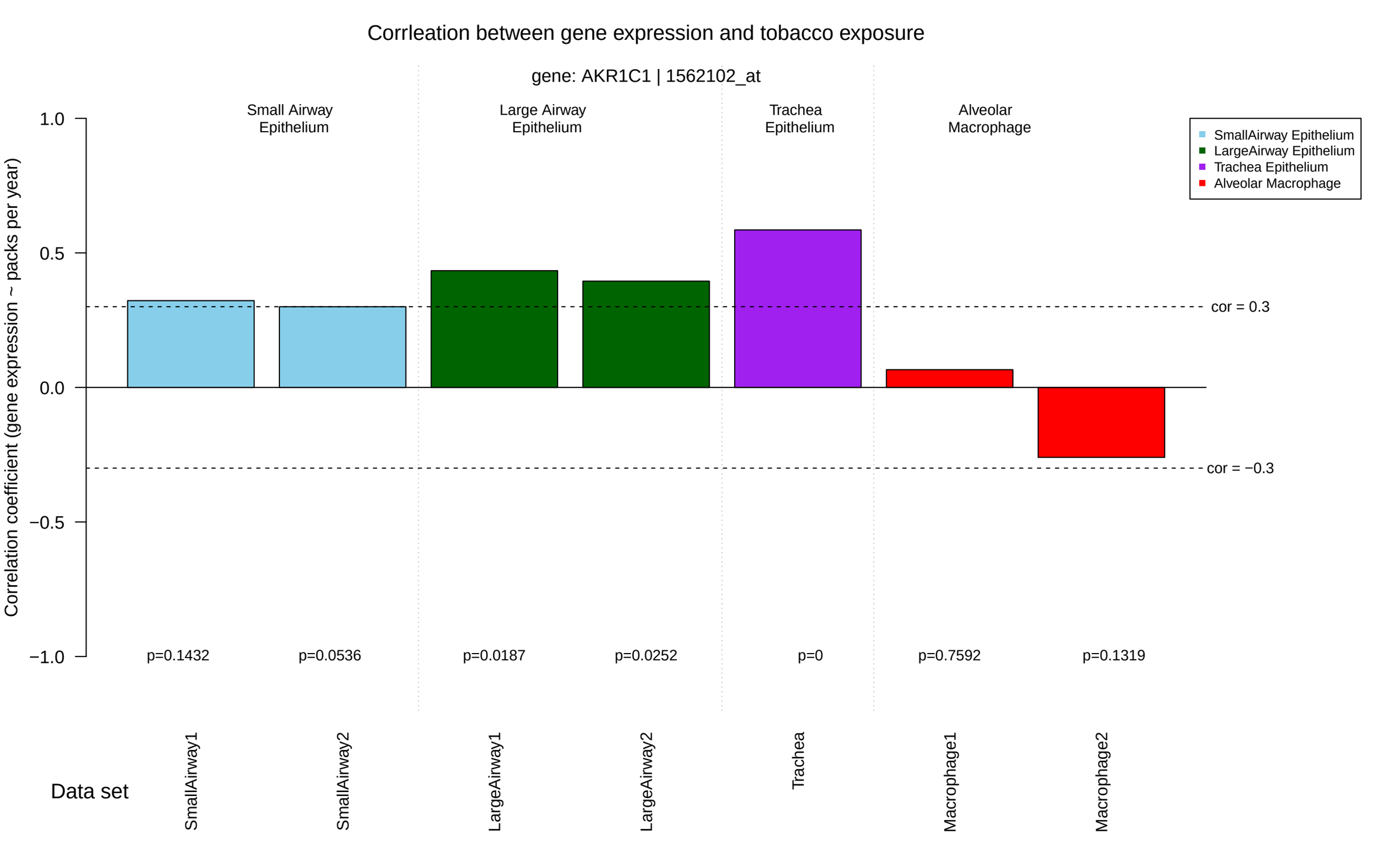

set: LargeAirway1 (p=6e−04)
AKR1C1 | 1562102_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0395)
AKR1C1 | 1562102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=8e−04)
AKR1C1 | 1562102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7099)
AKR1C1 | 1562102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0691)
AKR1C1 | 1562102_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR1C1 | 1562102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1432
p=0.0536
p=0.0187
p=0.0252
p=0
p=0.7592
p=0.1319
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2